Unit-5 Answer Key Microcontroller programming & Application.

STEPPER MOTOR CONTROL USING 8051

These Motors are called stepping motors (or) stop motors used for position control application such as disk drives and robotics.

→ The common stepper motor have 4 stator winding which are paired with a centre-tapped common Terminal.

→ The stepper motor shaft moves in a fixed repeatable increment, which allows one to move it to a precise position. The typical stepper motor has 50 teeth on rotor and eight poles on stator for a 1.8° stop angle.

Stop angle :-

The angle through which the motor shaft rotate for each Command Pulse (given by Mc) is called stop angle

Common stop angles are 1.8°, 2.5°, 7.8°, 15°.

$$B = \frac{Ns - Nr}{Ns \times Nr}$$

$$\text{Stop / sec} = \frac{\text{RPM} \times \text{steps perrevolution}}{60^\circ}$$

APPLICATION :- Printers, Textile industry, computers, robotics.

steps perrovolution.

→ Total no. of Steps needed to Complete one revolution (or) 360°

The swintching sequence shaum above is called 4 step switching sequence. since after four steps the same two swinding will be "ON".

stepper mator control using microcontroller:-

Draw the diagram using microcontroller

Here, the common wires are conraled to the positine four side of the motors power supply. Here +5V in suffilent. The four leads of the stator condoling are controled by four bits of the 8051 poot ie, p1.0, p1.1, p1.2, p1.3. The driver such as ULN 2003 to Energize the stator wondings. we can even use transistors as drivers instead of ULN 2003, as shown below.

Draw schematic diagram

Construction: stepper mator has a Permanent magnetic rolar (called) surrounded by station. The mast stepper Motor have four station windings that are paired with centre tapped Common.

The center tap allows a change of current direction in each of two coils when winding is grounded, thereby resulting in polarity change of the station. the fixed Movement is possible as a result of basic magnetic Theory where poles of the Same polarity repel and opposite poles attract. The direction of rotation is dictated by stator poles. the stator poles are determined by the current sent through the wire. Coils. As the direction of current is changed, the polarity is also changed causing the reverse rotation of the rotor. The stepper mator has four station windings and common for the center tapped leads. As the sequence of power is applied to each station winding the rotor will rotate. The normal four sequences are shown below:

Sequence diagram

If transistors are used, diods are also used to take care of the Inductive current generated when the coil is turned off. But if ULN 2003 is used it has internal diode to take care of back emf.

Algorithm:

1. Load the step sequence to accumulator.
2. Initialize ports (here port)
3. Move the step count for 4 steps sequence from accumulator. o Part rotate right clock wise.
4. call delay

CLOCK WISE :-

START: Mov 30, #03H : store the excitation value clock wise ratation.

Mov 31 , #06 H

MOV 32 , # OCH

MOV 33 , # 09H

RPT: Mov Ro, # 80H : initialize the memory pointer.

Mov R1 #04H

Loop: Mov P0, @R0

Lcall DELAY ; call subroutine

INC R0 ; Point next memory location.

DJNZR, Loop; Decrement next Counter, if not zero get next data.

SJMP RPT ; Loop again for notation.

DELAY : Mov R3, # Count; Load Register R3 with Count value.

HERE : DJNZ R3, HERE

RET ; Return to subroutine.

ANTICLOCKWISE ROTATION

START: Mov 30, # 03H ; store the excitationg values for bi-value phase clock wise ratation internal monory.

Mov 31, # 06H

Mov 32, # 0CH

Mov 33, # 09H

RPT : # 04R0, H33H

Mov R1, # 0HH

Loop :Mov P0, @ R0

Lcall DELAY

DEC R0

DJNZ R1, Loop

SJMP RPT

DELAY : MOV R3 , # Count

HERE : DJNZ R3 , HERE

RET

Program to ratate the stepper mator 90° clockwise and 180° anti-clockwise and then continuously repeat the seme sequence.

```
START : MOV R0 , # 32H    ; Initialize counter for 90° rotation.
MOV A , # 00010001B
Loop1 : MOV P0 . A
RLA
LCALL DELAY
DJNZ R0 , LOOP1
MOV R0 , # 64H
MOV A , # 00010001B
Loop 2 : MOV P0 , A
RRA
DJNZ R0 Loop 2
SJMP START.
DELAY : MOV R3 , # Count
HERE : DJNZ R3 , HERE
RET.
```